Reading

Fundamentals of Electric Circuits 5th Edition, McGrawHill, 2015
By Charles K. Alexander and Matthew N. O. Sadiku

- Chapter 1
- Chapter 2
- Chapter 6

Applications









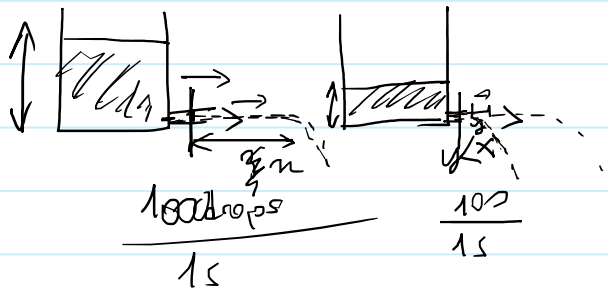




Charge and Current



$$i \triangleq \frac{dq}{dt} \quad \longleftrightarrow \quad Q \triangleq \int_{t_0}^{t} i\,dt$$



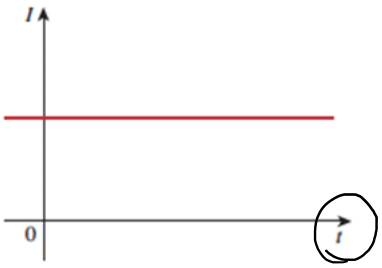
1 Coulomb of electrons ≈ 6,241,509,074,460,762,607 electrons

6×10^{18}

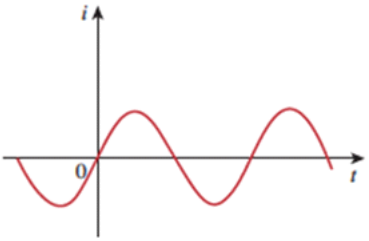
$$i = \frac{\Delta Q}{\Delta t}$$

$$\frac{1\,\text{Coulomb}}{1\,s} = 1\,A$$

DC



AC



Voltage



$$v_{ab} \triangleq \frac{dw}{dq}$$

1 volt = 1 joule/coulomb = 1 newton-meter/coulomb

Power

$$p \triangleq \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = \frac{dw}{dt}$$

V (dw/dq) · I (dq/dt)

V . I

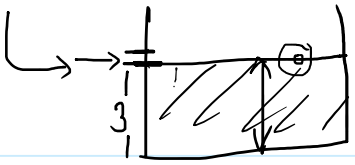
$$\int dw = \int p \cdot dt$$

$$w = \int p\, dt$$

$$w = \int_{t_0}^{t} p\, dt = \int_{t_0}^{t} vi\, dt$$

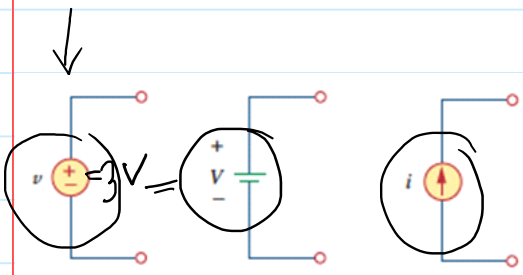
$$\sum p = 0$$

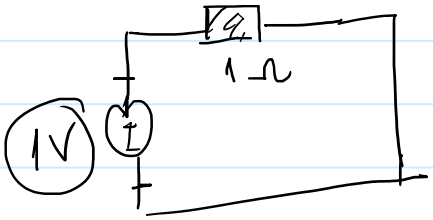
Active Circuit Elements

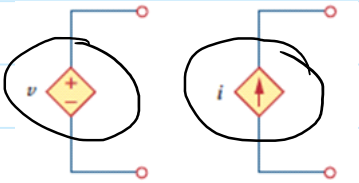


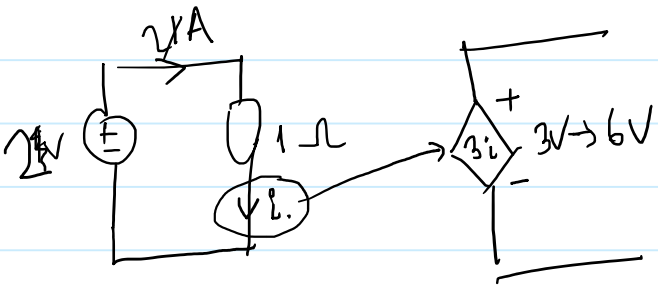


3 drop/sec ⇐⇒ 3A

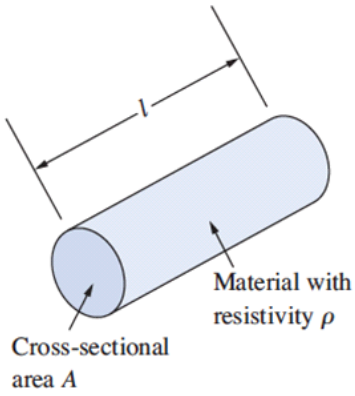


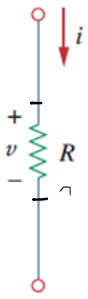






Ohm's Law



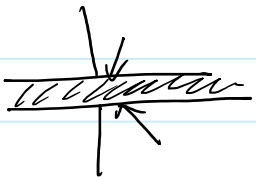


$$v = iR$$

$$R = \frac{l \cdot \rho}{A}$$

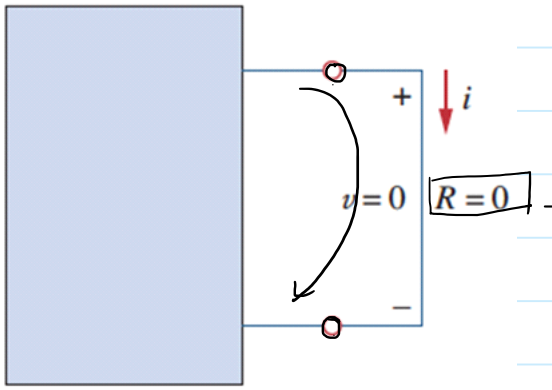
Resistivities of common materials.

Material	Resistivity (Ω·m)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^{2}	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

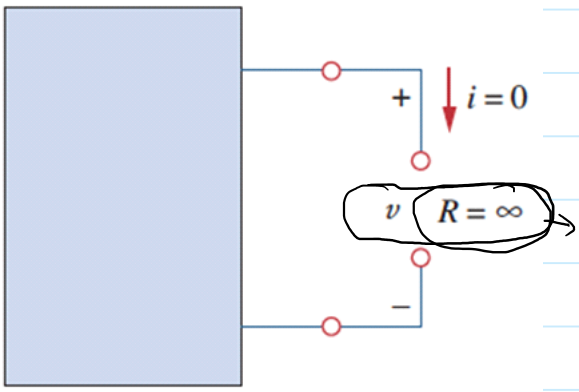


Short and Open Circuit

$v = iR \to 0$



$= 0.00001\ \Omega$

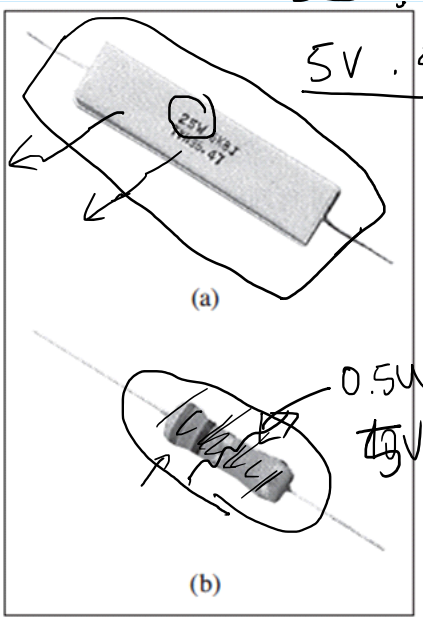


$v = iR$

$0 \quad \uparrow \quad 0 \quad \uparrow \to \infty$

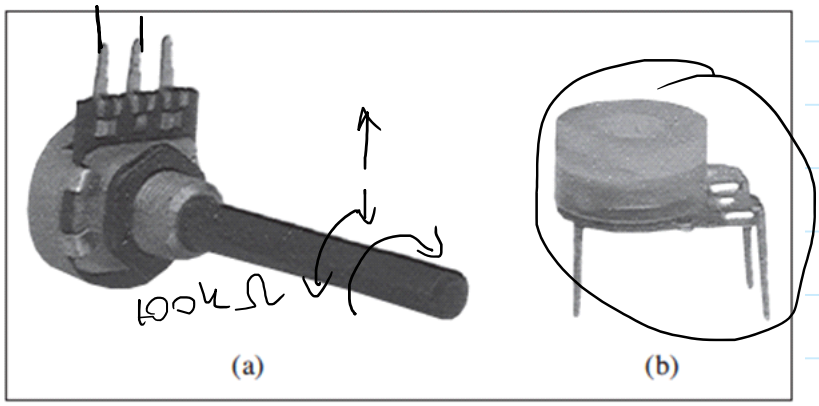
$v < 1000$

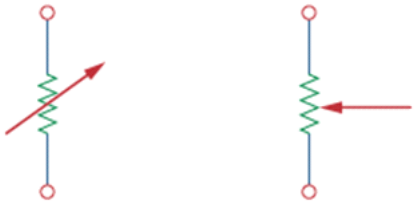
Types of Resistors



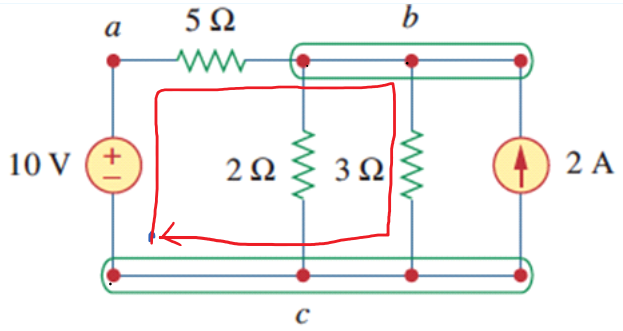
SMD







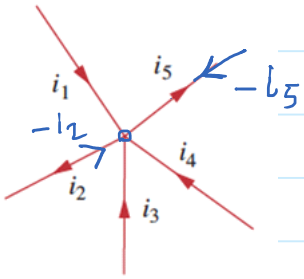
Branch, Node and Loop



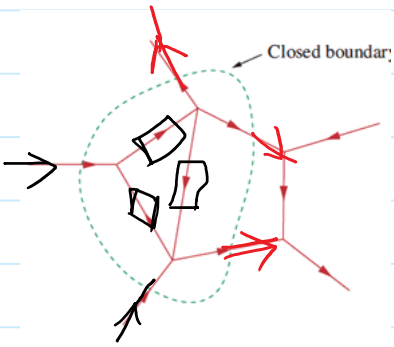
Kirchhoff's Laws - KCL

$$Q = \int i \, dt$$

$$\int i_1 + i_4 + i_3 \, dt = \int i_2 + i_5 \, dt$$

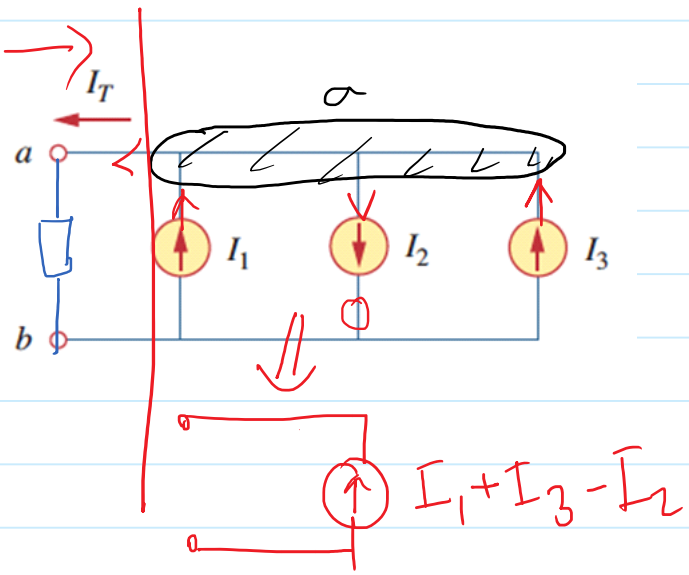


$$\sum_{n=1}^{N} i_n = 0$$



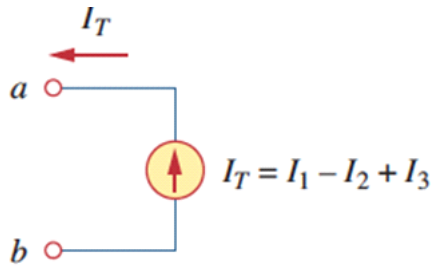
KCL

Kirchhoff's Laws - KCL



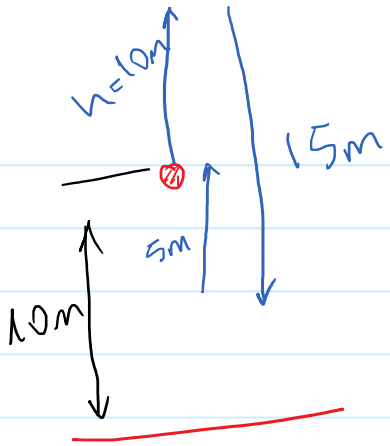
$$I_1 + I_3 = I_T + I_2$$

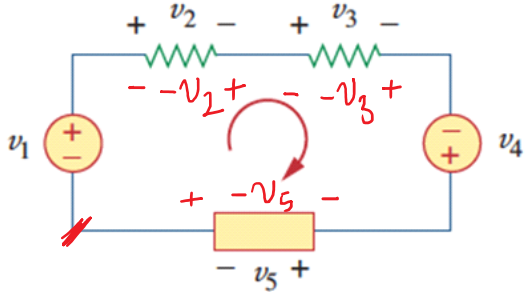
$$I_T = I_1 + I_3 - I_2$$



Kirchhoff's Laws - KVL

$$\sum_{m=1}^{M} v_m = 0$$

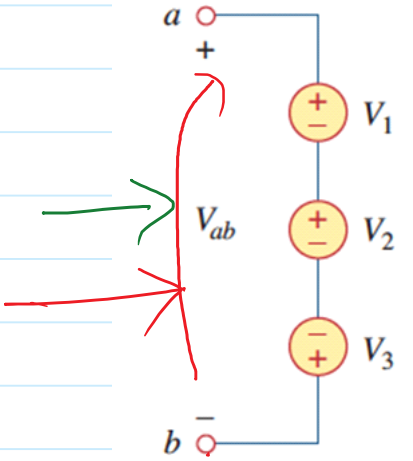




$$v_4 + v_1 = v_2 + v_3 + v_5$$

$$0 = v_1 + (-v_2) + (-v_3) + v_4 + (-v_5)$$

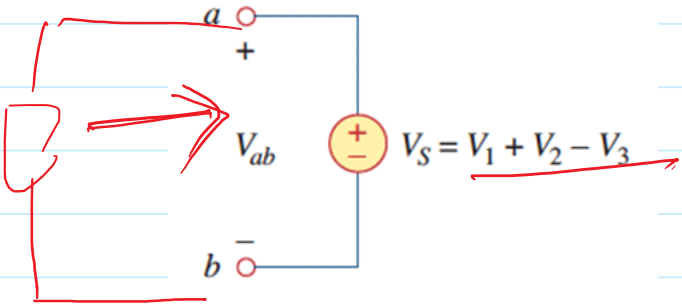
Kirchhoff's Laws - KVL



$-V_1$
$+$
$-V_2$
$+$
$-$
$+V_3$

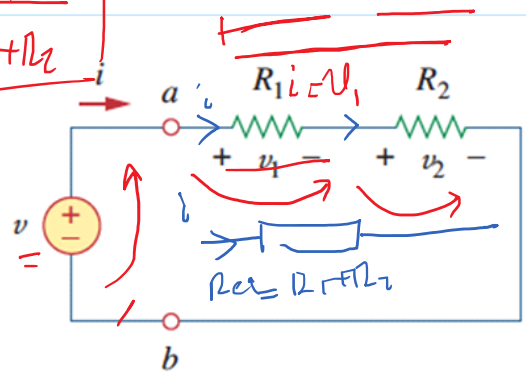
$V_{ab} + (-V_1) + (-V_2) + V_3 = 0$

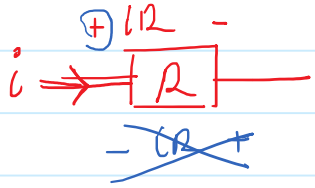
$V_{ab} = V_1 + V_2 - V_3$



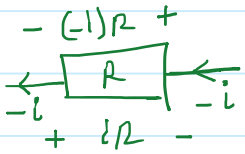
Series Resistors and Voltage Division

$$v_1 = v\frac{R_1}{R_1+R_2}$$

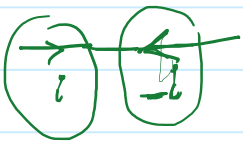




$v = iR$



$R_{eq} = R_1 + R_2$



$v + (-v_1) + (-v_2) = 0$

$v = v_1 + v_2$

$R_{eq} = R_1 + R_2$

$v = iR_1 + iR_2 = i(R_1 + R_2)$

$i = \frac{v}{R_1+R_2}$

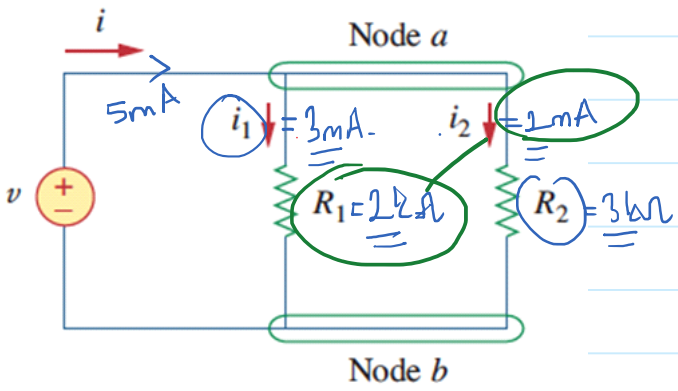
$$v_1 = \frac{R_1}{R_1 + R_2}v, \qquad v_2 = \frac{R_2}{R_1 + R_2}v$$

$v_1 = R_1 \cdot i = R_1 \frac{v}{R_1+R_2}$

$v_1 = v \cdot \frac{R_1}{R_1+R_2}$

$$R_{eq} = R_1 + R_2 + \cdots + R_N = \sum_{n=1}^{N} R_n$$

Parallel Resistors and Current Division

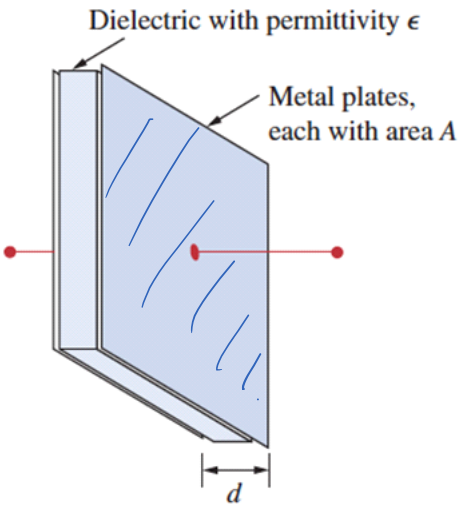


$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

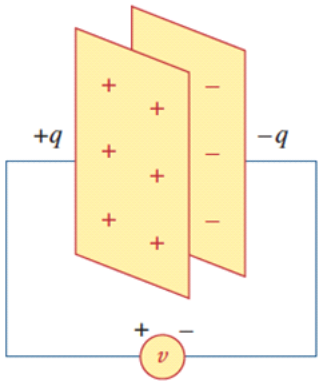
$$i_1 = \frac{R_2\, i}{R_1 + R_2}, \qquad i_2 = \frac{R_1\, i}{R_1 + R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$$

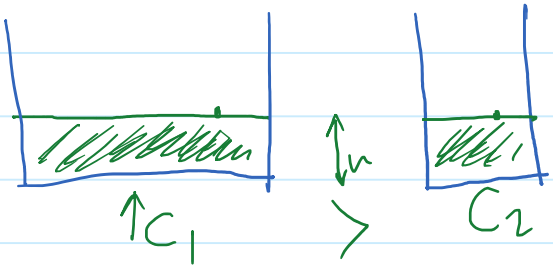
Capacitance



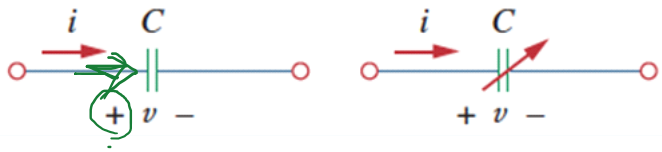
$$C = \frac{\epsilon A}{d}$$



$$q = Cv$$

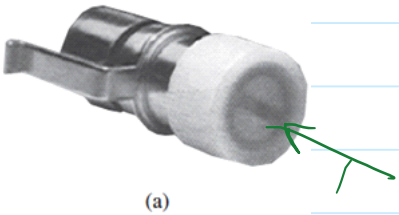


Types of Capacitors

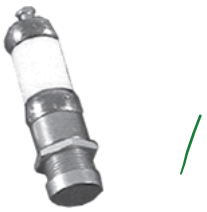


$$i = C\frac{dv}{dt}$$





(a)



(b)

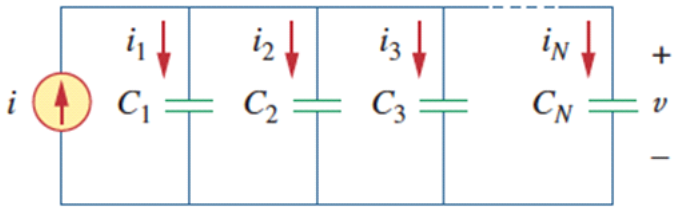
Capacitors

$$q = Cv$$

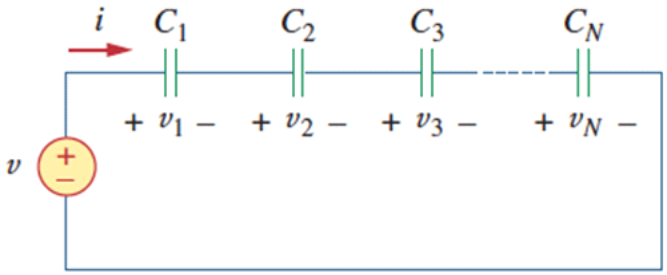
$$i = C\frac{dv}{dt}$$

$$w = \int_{-\infty}^{t} p(\tau)\,d\tau = C\int_{-\infty}^{t} v\frac{dv}{d\tau}d\tau = C\int_{v(-\infty)}^{v(t)} v\,dv = \frac{1}{2}Cv^2\Big|_{v(-\infty)}^{v(t)}$$

$$w = \frac{1}{2}Cv^2$$

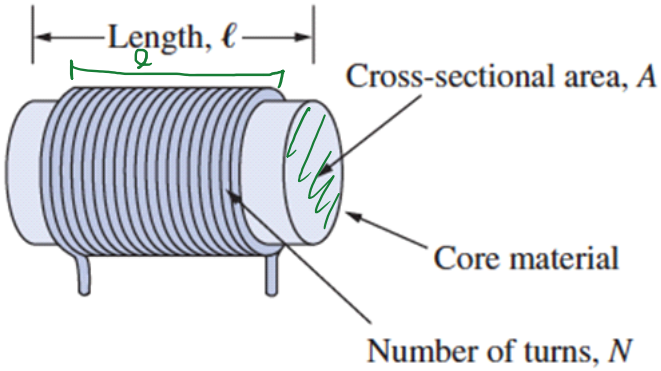


$$C_{eq} = C_1 + C_2 + C_3 + \cdots + C_N$$



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots + \frac{1}{C_N}$$

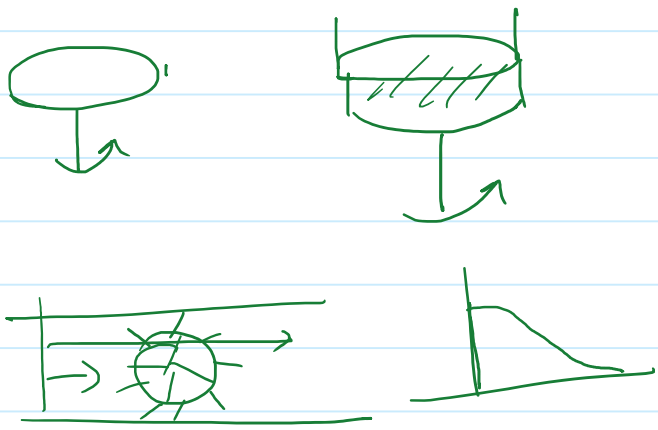
Inductance



$$L = \frac{N^2 \mu A}{\ell}$$

$$v = L\frac{di}{dt}$$

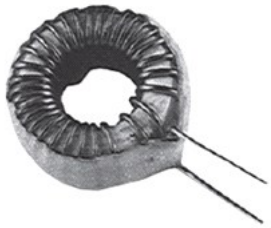
$$i = \frac{C dv}{dt}$$



Types of Inductors



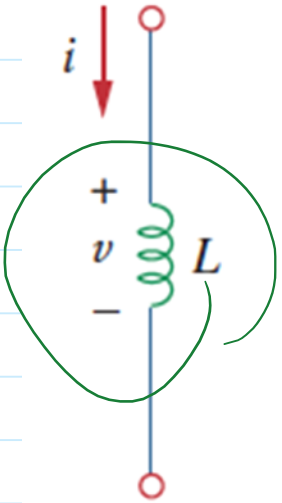
(a)



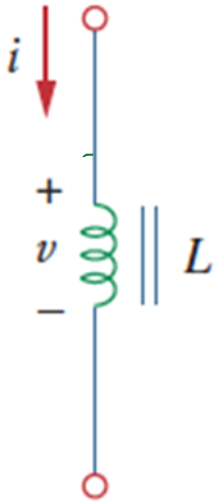
(b)



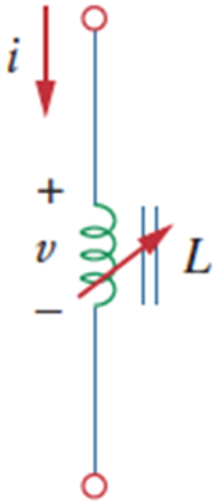
(c)



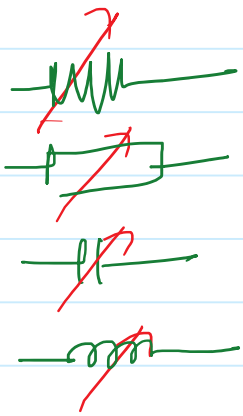
air-core



iron-core



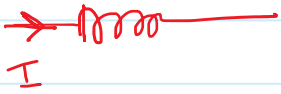
variable iron-core



Inductor

$V = Ri$

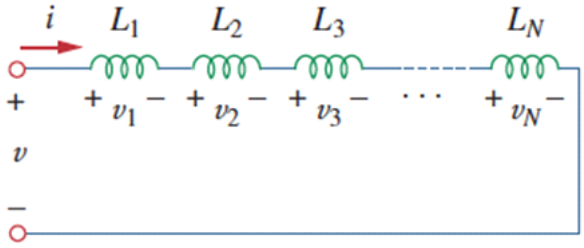
$$v = L\frac{di}{dt}$$



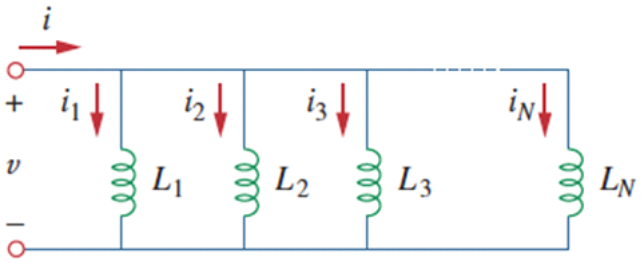
$\frac{1}{2} I^2 L$

$$w = \int_{-\infty}^{t} p(\tau)\,d\tau = L\int_{-\infty}^{t} \frac{di}{d\tau}\, i\,d\tau = L\int_{-\infty}^{t} i\,di = \frac{1}{2}Li^2(t) - \frac{1}{2}Li^2(-\infty)$$

$$w = \frac{1}{2}Li^2$$



$$L_{eq} = L_1 + L_2 + L_3 + \cdots + L_N$$



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \cdots + \frac{1}{L_N}$$

Summary

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i